REV 2026 | Practicum | Semester II | Credits 4.5 | Contact Hours 90

Chemistry for Technical Practices

Course Code: 2003B · Polytechnic Diploma (Polymer Technology, Printing Technology, Textile Technology, Chemical Engineering)

Programme(s)	Polymer Technology, Printing Technology, Textile Technology, Chemical Engineering
Subject Code	2003B
Semester	II
Course Type	Practicum
Teaching Scheme	L:T:P:R = 3:0:3:0
Contact Hours	90
Credits	4.5
CIE Marks	40
ESE Marks	60

📜 Official Source Declaration

This handbook is based on the Kerala Polytechnic **Revision 2026** syllabus for **Course 2003B – Chemistry for Technical Practices**.

> **Verified inconsistencies in source PDF:** The syllabus table for Module IV in the "Detailed Syllabus" section lists "Experiments based on estimation of metals in alloys and preparation of polymers" under both Module IV and Module I subtopics. The handbook treats this as a typographical duplication and places the experimental content under the correct module (Module IV) as per the overall subject flow. Additionally, the CO-PO mapping shows partial correlations; all listed values are reproduced as given.

All content is elaborated for educational purposes; the syllabus is the authoritative source.

📊 Course Dashboard

4
Modules

90
Total Contact Hours

90
Self-Learning Hours

4.5
Credits

Teaching and Assessment Scheme

Component	Details

Component	Details
Teaching Scheme	L:T:P:R = 3:0:3:0
Contact Hours	90
Self-Learning Hours	90 (as per weekly activities)
Credits	4.5
CIE Marks	40
ESE Marks	60
Assessment Components	Attendance, CA1–CA5 (Self-learning, practical tests, series exams), ESE (Written + Lab)

📘 How to Use This Handbook

1. Understand

Read each module's concepts and explanations. Use the learning goals and key facts to focus.

2. Visualise

Study the labelled diagrams and tables. Use the Visual Library for quick reference.

3. Apply

Work through the examples, calculators, and numerical problems. Test yourself with module-wise Q&A.

4. Revise

Use the Quick Revision cards, formula bank, glossary, and model paper before exams.

💡 **Tip:** Use the **Print / Save as PDF** button to generate a printable PDF for offline study.

🎯 Course Objectives & Outcomes

Objectives

1. To impart knowledge about fundamental concepts in chemistry including atoms, molecules, solutions, thermodynamics and relate them to engineering applications.
2. To apply key concepts of analytical chemistry and water treatment methods for solving real-world domestic and industrial problems.
3. To impart basic knowledge about organic functional groups, dyes and pigments.
4. To familiarize students with various engineering materials for selecting appropriate materials in industrial and domestic applications and understand green chemistry principles.

Course Outcomes (CO)

Course Outcomes and Bloom's Taxonomy Level

CO	Course Outcome	Bloom's Level
CO1	Apply the basic concepts of atoms, molecules, solutions and thermodynamic processes in scientific and engineering contexts.	Apply
CO2	Apply the fundamentals of analytical Chemistry, appropriate water treatment methods and also the basic concepts of acids and bases to solve engineering problems.	Apply
CO3	Examine the structure, classification, and applications of dyes, pigments and organic functional groups through theoretical and experimental approaches.	Apply
CO4	Explore various engineering materials for suitable industrial and domestic applications and explain green Chemistry principles promoting sustainable practices.	Apply

CO-PO Mapping (as per syllabus)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	-
CO2	3	3	2	3	2	2	2	-	-	-	-
CO3	2	2	2	2	2	-	-	-	-	-	-
CO4	3	2	3	2	2	2	3	-	-	-	-

Legends: 1-Low; 2-Medium; 3-High; "-" No Correlation.

🗺️ Curriculum Coverage Map

Module → Topics → Hours → CO

Module	Topics	Hours	CO
I	Atoms, Molecules, Solutions, Thermodynamics, Preparation of standard solutions, Endothermic/Exothermic reactions	~22	CO1
II	Volumetric analysis, Hardness of water, Water treatment, pH/pOH, Quantitative/Qualitative analysis	~24	CO2
III	Organic functional groups, Dyes (classification, dye-fibre interactions), Pigments, Esterification, tests for unsaturation/aromaticity	~16	CO3
IV	Alloys, Polymers, Nanomaterials, MOFs, Biomaterials, Biofuel, Green Chemistry, Estimation of metals in alloys, preparation of polymers	~28	CO4

🧭 Module Roadmap

⚛️ Module I

Fundamentals: atoms, molecules, solutions, thermodynamics. Builds foundation for all chemical engineering applications.

🧪 Module II

Analytical chemistry & water treatment. Directly applicable to domestic/industrial problem-solving.

🎨 Module III

Organic chemistry, dyes, pigments. Critical for textile, printing, and polymer industries.

🧬 Module IV

Engineering materials (alloys, polymers, nanomaterials, biomaterials) and green chemistry. Integrates sustainability.

MODULE I

Atoms, Molecules, Solutions & Thermodynamics

📘 Contact Hours: ~22 🎯 CO1 📊 Bloom: Apply

This module covers the basic building blocks of matter, concentration expressions, and the laws governing energy changes — essential for all engineering chemistry.

📌 Learning Goals

- Define atom, molecule, solute, solvent.
- Calculate fundamental particles from atomic number/mass number.
- Solve molarity and normality problems.
- Explain thermodynamic systems, processes, and state functions.

🔑 Key Facts

- Z = protons = electrons (neutral atom).
- Molarity (M) = moles solute / L solution.
- Normality (N) = gram-equivalents / L solution.
- Enthalpy (H) = U + PV; Entropy (S) = disorder.

🔬 Atoms and Molecules

Atom: The smallest particle of an element that retains its chemical properties. Example: Hydrogen atom (H), Oxygen atom (O).

Molecule: A group of two or more atoms held together by chemical bonds. Example: H_2O (water), O_2 (oxygen gas).

Fundamental particles: Electron (e^-, charge -1, mass ~9.1×10^{-31} kg), Proton (p^+, charge +1, mass ~1.67×10^{-27} kg), Neutron (n^0, charge 0, mass ~1.67×10^{-27} kg).

Atomic number (Z): Number of protons in the nucleus. **Mass number (A):** Total number of protons + neutrons.

> **Example:** For Calcium (Ca, Z=20, A=40): protons=20, electrons=20, neutrons=40-20=20.

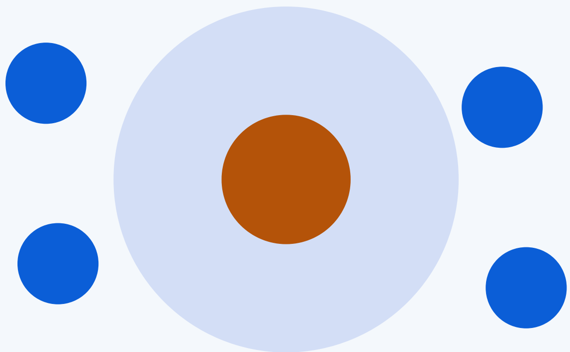

Fig: Schematic atom showing nucleus (protons+neutrons) and orbiting electrons.

> **Malayalam support:** □□□□□□ (atom) □□□□□□□□□□□□□□□ □□□□□□□□ □□□□□□ □□□□□□□□□□. □□□□□□□□□□ (molecule) □□□□□□ □□□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□. □□□□□□□□□□□, □□□□□□□□□□□, □□□□□□□□□□□ □□□□□□□□□□□□ □□□□□□□□□□ □□□□□□□.

🧪 Solutions and Concentration Expressions

Solute: Substance that dissolves (e.g., salt). **Solvent:** Substance that dissolves the solute (e.g., water). **Standard solution:** A solution of known concentration.

Molarity (M) = moles of solute / volume of solution (L)

Normality (N) = gram-equivalents of solute / volume of solution (L)

ppm = (mass of solute / mass of solution) × 10^6

> **Worked example (Molarity):** 58.5 g of NaCl (M.W. = 58.5 g/mol) dissolved in water to make 1 L solution. M = 58.5/58.5 / 1 = 1 M.

🧮 Molarity Calculator

Calculate molarity from mass, molecular weight, and volume.

Mass (g): 58.5

Molecular weight (g/mol): 58.5

Volume (L): 1

Molarity = 1.00 M

🔥 Thermodynamics: Systems, Processes, State Functions

System: The part of the universe under study. **Surroundings:** Everything else.

Types of systems: Open (mass & energy exchange), Closed (energy only), Isolated (neither).

Processes: Isothermal (constant T), Isobaric (constant P), Isochoric (constant V), Adiabatic (no heat exchange).

Reversible vs. Irreversible: Reversible proceeds infinitesimally slowly; irreversible is spontaneous.

State functions: Internal energy (U), Enthalpy ($H = U + PV$), Entropy (S).

Comparison of Thermodynamic Processes

Process	Constant	Example
Isothermal	Temperature	Slow expansion of gas in contact with a heat bath
Isobaric	Pressure	Heating gas in a cylinder with a movable piston
Isochoric	Volume	Heating gas in a rigid container
Adiabatic	No heat exchange	Rapid compression of gas in a insulated cylinder

> **Tip:** Endothermic reactions absorb heat ($\Delta H > 0$); exothermic release heat ($\Delta H < 0$).

🧪 Practical: Preparation of Standard Solutions & Endothermic/Exothermic Reactions

Preparation of 0.1 N oxalic acid solution: Weigh 6.3 g of oxalic acid ($H_2C_2O_4 \cdot 2H_2O$) and dissolve in distilled water to make 1 L.

Endothermic reaction: Citric acid + baking soda (absorbs heat, feels cold).

Exothermic reaction: Neutralisation HCl + NaOH (releases heat, feels warm).

> **Safety:** Wear gloves and goggles when handling acids and bases. Use a fume hood if required.

MODULE II

Analytical Chemistry & Water Treatment

📘 Contact Hours: ~24 🎯 CO2 📊 Bloom: Apply

Understand volumetric analysis, water hardness, treatment methods, pH concepts, and their application in solving engineering problems.

📊 Volumetric Analysis

Titration: A technique where a solution of known concentration (titrant) is added to a solution of unknown concentration (analyte) until the reaction is complete (end point).

Normality equation: $N_1V_1 = N_2V_2$

> **Example:** 20 mL of 0.1 N HCl neutralises 40 mL of NaOH. Find normality of NaOH.
> $N_1V_1 = N_2V_2 \rightarrow 0.1\times20 = N_2\times40 \rightarrow N_2 = 0.05$ N.

Choice of Indicators

Titration type	Indicator	pH range
Strong acid – Strong base	Phenolphthalein / Methyl orange	8.2–10 / 3.1–4.4
Strong acid – Weak base	Methyl orange	3.1–4.4
Weak acid – Strong base	Phenolphthalein	8.2–10

💧 Hardness of Water & Treatment Methods

Hard water: Contains dissolved salts of Ca^{2+}, Mg^{2+} (bicarbonates, chlorides, sulphates). **Temporary hardness:** $Ca(HCO_3)_2$, $Mg(HCO_3)_2$ (removed by boiling). **Permanent hardness:** $CaCl_2$, $MgCl_2$, $CaSO_4$, $MgSO_4$ (removed by chemical treatment).

Municipal water treatment steps: Screening → Sedimentation → Coagulation → Filtration → Sterilisation.

Sterilisation methods: Chlorination, UV radiation, Ozonation.

Advanced treatment: Reverse osmosis, Nano-filtration.

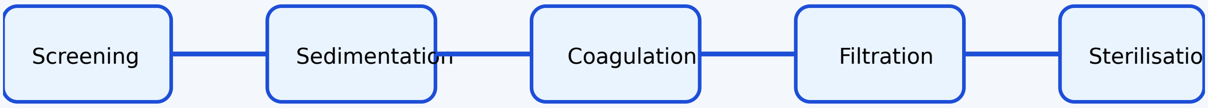

Fig: Municipal water treatment flow chart.

🧪 Acids, Bases, pH and pOH

Arrhenius concept: Acid produces H^+ in water; base produces OH^- in water.

$pH = -\log_{10}[H^+]$, $pOH = -\log_{10}[OH^-]$, $pH + pOH = 14$ (at 25°C).

> **Example:** $[H^+] = 1\times10^{-4}$ M → pH = 4 (acidic). $[OH^-] = 1\times10^{-3}$ M → pOH = 3 → pH = 11 (basic).

🧮 pH Calculator

$[H^+]$ (M): 1e-4

pH = 4.00

🧪 Practical: Volumetric Analysis & pH Determination

- Standardisation of HCl using Na_2CO_3.
- Estimation of NaOH using standard HCl.
- Estimation of KOH using standard oxalic acid.
- Standardisation of EDTA using $ZnSO_4$.
- Volumetric estimation of total hardness of water using EDTA.
- Determination of pH using pH meter, universal indicator, pH paper.

MODULE III

Organic Functional Groups, Dyes & Pigments

📘 Contact Hours: ~16 | 🎯 CO3 | 📊 Bloom: Apply

Explore the structure, classification, and applications of organic functional groups, dyes, and pigments — essential for textile and printing technologies.

🧪 Organic Functional Groups

Functional group: A specific group of atoms within a molecule that is responsible for its characteristic chemical reactions.

- **Alcohol:** -OH (e.g., ethanol)
- **Acid:** -COOH (e.g., acetic acid)
- **Ester:** -COOR (e.g., ethyl acetate)
- **Hydrocarbons:** Aliphatic (linear/branched) and Aromatic (benzene ring); Saturated (single bonds) and Unsaturated (double/triple bonds).

Examples of Functional Groups

Group	Structure	Example
Alcohol	-OH	Ethanol (CH_3CH_2OH)
Carboxylic acid	-COOH	Acetic acid (CH_3COOH)
Ester	-COO-	Ethyl acetate ($CH_3COOCH_2CH_3$)

🎨 Dyes: Classification, Chromophore, Auxochrome

Dye: A coloured substance that imparts colour to a material.

Chromophore: The part of the molecule responsible for colour (e.g., -N=N-, $-NO_2$).

Auxochrome: A group that intensifies colour and helps in bonding (e.g., -OH, $-NH_2$).

Classification by source: Natural (e.g., indigo) and Synthetic (e.g., azo dyes).

Classification by application: Direct, Acid, Basic, Disperse, Mordant, Vat dyes.

Classification by structure: Azo, Anthraquinone, Nitro, Indigo dyes.

Dye-fibre interactions: Physical adsorption, Hydrogen bonding, Ionic, Covalent interactions.

> **Tip:** Direct dyes are applied directly to cotton; vat dyes are insoluble and require reduction.

🖌️ Pigments

Pigment: A substance that imparts colour by reflecting light; usually insoluble and dispersed in a medium.

Types: Inorganic (e.g., Titanium dioxide, Iron oxide) and Organic (e.g., Phthalocyanine).

Uses: Paints, inks, plastics, cosmetics, textiles.

🧪 Practical: Preparation of Dyes, Esterification, Identification Tests

- **Preparation of azo dye:** Diazotisation and coupling reaction.
- **Esterification reaction:** Alcohol + carboxylic acid → ester + water (with acid catalyst).
- **Identification tests:** Alcohol (ceric ammonium nitrate test), Acid (litmus test), Ester (hydroxamic acid test).
- **Test for unsaturation:** Baeyer's test ($KMnO_4$ decolourisation).
- **Test for aromaticity:** Baeyer's test (resistance to $KMnO_4$) and ignition test (sooty flame).

MODULE IV

Engineering Materials & Green Chemistry

📘 Contact Hours: ~28 🎯 CO4 📊 Bloom: Apply

Study the properties, applications, and sustainability aspects of alloys, polymers, nanomaterials, biomaterials, and green chemistry.

🔩 Alloys

Alloy: A mixture of two or more metals (or a metal and a non-metal).

Purposes of alloying: To increase strength, corrosion resistance, hardness, and to improve appearance.

Common Alloys and Applications

Alloy	Composition	Applications
Stainless steel	Fe + Cr + Ni	Kitchenware, surgical instruments
Brass	Cu + Zn	Decorative items, plumbing fittings
Bronze	Cu + Sn	Statues, bearings
Solder	Pb + Sn	Joining electrical wires

🧵 Polymers

Monomer: Small repeating unit. **Polymer:** Large molecule made of many monomers. **Polymerisation:** The process of forming polymers.

Homo vs Copolymer: Homopolymer: one type of monomer (e.g., PE). Copolymer: two or more types (e.g., Buna-S).

Thermoplastics: Soften on heating (e.g., PE, PVC, Nylon-66). **Thermosetting plastics:** Set permanently on heating (e.g., Bakelite).

Common Polymers and Applications

Polymer	Monomer	Applications
PE	Ethylene	Plastic bags, containers
PVC	Vinyl chloride	Pipes, cables
Nylon-66	Hexamethylene diamine + Adipic acid	Fibres, ropes
Bakelite	Phenol + Formaldehyde	Electrical switches, handles
PET	Ethylene glycol + Terephthalic acid	Bottles, fibres
PU	Diisocyanate + Polyol	Foams, coatings
PMMA	Methyl methacrylate	Transparent sheets, lenses

Natural rubber: Monomer is isoprene. **Vulcanisation:** Cross-linking with sulphur to improve strength, elasticity, and heat resistance.

Synthetic rubber: Buna-S (Styrene + Butadiene), Buna-N (Acrylonitrile + Butadiene).

🧬 Nanomaterials, MOFs, Biomaterials, Biofuel, Green Chemistry

Nanomaterials: Materials with at least one dimension in the 1–100 nm range. Examples: Carbon nanotubes (CNTs), Fullerenes (C_{60}), Graphene (single layer of carbon). Applications: Electronics, medicine, composites.

Metal-Organic Frameworks (MOFs): Porous materials made of metal ions and organic linkers. Applications: Gas storage, catalysis, drug delivery.

Biomaterials: Materials used in biomedical applications. Types: Metallic (e.g., Ti implants), Ceramic (e.g., bone grafts), Polymeric (e.g., sutures), Composite (e.g., dental fillings).

Biofuel: Fuel produced from biological sources (e.g., ethanol from corn, biodiesel from vegetable oils). Applications: Transportation, power generation.

Green Chemistry: Design of chemical products and processes that reduce or eliminate hazardous substances. **Principles:** Prevention, Atom economy, Less hazardous synthesis, Safer solvents, Energy efficiency, Renewable feedstocks, Catalysis, etc.

🧪 Practical: Estimation of Metals in Alloys & Polymer Preparation

- Estimation of copper in brass.
- Estimation of zinc in brass.
- Estimation of iron in iron ore.
- Preparation of Urea-formaldehyde resin.
- Preparation of Phenol-formaldehyde resin.

📐 Formula Bank

Molarity
M = moles solute / L solution

Normality
N = gram-equivalents / L solution

ppm
(mass solute / mass solution) × 10^6

Normality equation
$N_1V_1 = N_2V_2$

pH
$pH = -\log_{10}[H^+]$

pOH
$pOH = -\log_{10}[OH^-]$

pH + pOH
= 14 (at 25°C)

Enthalpy
$H = U + PV$

Entropy
S = disorder measure

📊 Visual Library

⚛️ Atom Model

See Module I – Nucleus with protons/neutrons and orbiting electrons.

💧 Water Treatment Flow

See Module II – Screening → Sedimentation → Coagulation → Filtration → Sterilisation.

🎨 Dye-Fibre Interactions

See Module III – Physical adsorption, H-bonding, ionic, covalent.

🧬 Polymer Types

See Module IV – Thermoplastics vs Thermosetting plastics.

📝 Self-Learning Activities

Module I

- Assignment: atomic number/mass number of 20 elements.
- Model of atom using chart/paper.
- Report on concentration terms.
- Observe endothermic/exothermic processes at home.

Module II

- Problems on normality equation.
- Natural indicators from household materials.
- Collect water samples and identify pH.
- Report on water purification methods.

Module III

- Report on natural and synthetic dyes in textile printing.
- Video on dyeing clothes.
- Short note on esters in daily life.
- Report on pigments used in textile/printing.

Module IV

- Assignment on articles made of alloys.
- Report on bioplastics.
- Short video on nanomaterials.
- Report on biomaterials and MOFs.
- Assignment on biofuels.

📋 Assessment Methodology

Assessment Components

Mode	Description	Marks	Converted to
Attendance	–	–	5
CA1	Average of Self-learning activities from each module	50	10
CA2	Practical test (first half)	40	10
CA3	Practical test (second half)	40	10
CA4	Series Exam (2 modules)	50	10
CA5	Series Exam (2 modules)	50	10
ESE	Written Exam (theory) + Lab Examination	60 + 40	60

Series Exam Pattern (2 hours)

Part A: 2 questions × 1 mark = 2
Part B: 3 questions × 3 marks = 9
Part C: 1 set (with choice) × 7 marks = 7
Total: 18 marks (per module, 2 modules combined)

ESE Pattern (2.5 hours)

Part A: 2 questions/module × 1 mark = 8
Part B: 2 out of 3/module × 3 marks = 24
Part C: 1 set with choice/module × 7 marks = 28
Total: 60 marks

Note: Minimum of 10 experiments from all modules must be performed.

❓ Module-wise Questions & Answers

Module I

1. Define atom and molecule with examples.

Atom is the smallest particle of an element that retains its properties (e.g., H). Molecule is a group of atoms bonded together (e.g., H_2O).

2. Calculate the number of protons, neutrons and electrons in an atom with Z=17, A=35.

Protons=17, electrons=17, neutrons=35-17=18.

3. What is molarity? Calculate the molarity of a solution containing 40 g NaOH (M.W. 40) in 2 L water.

M = moles/L = (40/40)/2 = 0.5 M.

4. Distinguish between isothermal and adiabatic processes.

Isothermal: constant temperature; Adiabatic: no heat exchange.

Module II

1. State the principle of volumetric analysis.

At the equivalence point, equivalents of acid = equivalents of base ($N_1V_1=N_2V_2$).

2. Differentiate between temporary and permanent hardness of water.

Temporary: caused by bicarbonates, removed by boiling. Permanent: caused by chlorides/sulphates, removed by chemical treatment.

3. Draw a flow chart for municipal water treatment.

Screening → Sedimentation → Coagulation → Filtration → Sterilisation.

4. Calculate pH of a solution with $[H^+] = 1\times10^{-6}$ M.

pH = $-\log(10^{-6}) = 6$.

Module III

1. Define chromophore and auxochrome with examples.

Chromophore: colour-producing group (e.g., -N=N-). Auxochrome: intensifies colour (e.g., -OH).

2. Classify dyes based on method of application.

Direct, Acid, Basic, Disperse, Mordant, Vat.

3. What are pigments? Give two examples.

Insoluble colouring substances; e.g., TiO_2 (inorganic), Phthalocyanine (organic).

4. Describe the test for unsaturation.

Baeyer's test: addition of $KMnO_4$ (purple) → decolourisation indicates unsaturation.

Module IV

1. Define alloy. List the purposes of alloying.

Alloy is a mixture of metals. Purposes: increase strength, corrosion resistance, hardness.

2. Differentiate between thermoplastics and thermosetting plastics.

Thermoplastics soften on heating (e.g., PE); thermosetting set permanently (e.g., Bakelite).

3. What is vulcanisation? List properties of vulcanised rubber.

Vulcanisation: cross-linking rubber with sulphur. Properties: stronger, more elastic, heat resistant.

4. State any three principles of green chemistry.

Prevention, Atom economy, Safer solvents.

📄 Practice Model Paper

Based on the official examination pattern. Not an official SITTTR model question paper.

Time: 2.5 hours | Max Marks: 60

Part A (1 mark each – 2 questions per module)

- 1. Define atomic number.
- 2. What is ppm?
- 3. State the normality equation.
- 4. What is the pH of pure water?
- 5. Define functional group.
- 6. What is a chromophore?
- 7. Define alloy.
- 8. What is vulcanisation?

Part B (3 marks each – 2 out of 3 per module)

- 1. Explain molarity and normality with formulas.
- 2. Distinguish between open and closed systems.
- 3. Explain the steps in municipal water treatment.
- 4. What is the choice of indicator in acid-base titrations?
- 5. Classify dyes based on chemical structure.
- 6. Explain the esterification reaction.
- 7. List the applications of common polymers.
- 8. Explain the principles of green chemistry.

Part C (7 marks each – 1 set with choice per module)

- 1. a) Calculate the number of particles in an atom with Z=20, A=40. b) Solve a molarity problem.
- 2. a) Explain hardness of water and its removal. b) Calculate pH from $[H^+]$.
- 3. a) Discuss the classification of dyes. b) Explain dye-fibre interactions.
- 4. a) Explain polymers and their applications. b) Discuss nanomaterials and green chemistry.

⚡ Quick Revision

Module I

- Z = protons = electrons (neutral)
- M = moles/L; N = eq/L
- $\Delta H = H(\text{products}) - H(\text{reactants})$

Module II

- $N_1V_1 = N_2V_2$
- $pH = -\log[H^+]$
- Water treatment: screening → sedimentation → coagulation → filtration → sterilisation

Module III

- Chromophore = colour; Auxochrome = intensifier
- Dye classification: application (direct, acid, etc.) and structure (azo, anthraquinone, etc.)

Module IV

- Alloying improves properties
- Thermoplastics vs Thermosetting
- Green Chemistry: 12 principles

⚠️ **Common mistakes:** Confusing molarity and normality; forgetting units in pH calculations; misclassifying dyes; not distinguishing between thermoplastics and thermosetting.

📖 Glossary

Term	Meaning
Atom	Smallest particle of an element
Molecule	Group of atoms bonded together
Molarity	Moles of solute per litre of solution
Normality	Gram-equivalents of solute per litre of solution

Term	Meaning
pH	Negative logarithm of H^+ concentration
Alloy	Mixture of metals
Polymer	Large molecule made of repeating monomers
Chromophore	Colour-producing group in a dye
Green Chemistry	Design of environmentally benign chemical processes

📚 References

Textbooks

Title	Author
Principles of Physical Chemistry	Puri, Sharma & Pathania
Vogel's Textbook of Quantitative Chemical Analysis	Arthur Israel Vogel
Fundamentals of Analytical Chemistry	Skoog, West, Holler
Nanoscience and Technology	Muralidharan & Subramania
Color Chemistry	Heinrich Zollinger

Web Resource: NPTEL – Chemistry